

Visvesvaraya Technological University
Belgaum, Karnataka- 590014



Third Year
A Mini-Project Report

On

**“ Explainable Multimodal Brain – Heart Stress Detection and
AI-Driven Intervention System using EEG and ECG Signals ”**

Submitted in the partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

Submitted by

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2025–2026

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Certificate

This is to certify that the Mini Project Work entitled **“Explainable Multimodal Brain – Heart Stress Detection and AI-Driven Intervention System using EEG and ECG Signals”** is a bonafide work carried out by **Ashlesh (1DS23CD012), Sambhram S Naregal (1DS23CD045), Tushar D Kudachimath (1DS23CD058) and Vishal Krishna Bhat K (1DS23CD063)** , in partial fulfilment for the VI semester of Bachelor of Engineering in CSE (Data Science) of the Visvesvaraya Technological University, Belgaum during the year 2025–2026. The Project report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

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1.

2.

ACKNOWLEDGEMENT

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Chapter 1

Introduction

Write the complete introduction of the project here.

Chapter 2

Literature Survey

Discuss existing systems and related research.

Chapter 3

Research Gap Identified

Explain the limitations in existing systems.

Chapter 4

Architectures

4.1 High Level Architecture

Insert high level architecture diagram here.

4.2 Detailed Architecture

Insert detailed architecture diagram here.

4.3 UML Diagrams

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram

Chapter 5

Project Objectives

- Objective 1
- Objective 2
- Objective 3
- Objective 4

Chapter 6

Timeline using Gantt

Insert Gantt chart here.

Chapter 7

Hardware and Software Requirements

7.1 Hardware Requirements

- Processor
- RAM
- Storage

7.2 Software Requirements

- Operating System
- Programming Language
- Frameworks
- Database

Chapter 8

Sample Code

Example code here

Chapter 9

Project Demo Screenshots

Insert screenshots here.

Chapter 10

AI Tools Application

Explain AI tools used in the project.

Chapter 11

SGDs Met by Project with Proper Justification

Mention SDGs achieved by the project.

Chapter 12

Purpose of the Project

Explain target audience and future scope.

Chapter 13

Research Paper Draft

Attach research paper draft and Turnitin report. Artificial Intelligence is transforming healthcare systems rapidly [2, 1].

References and Bibliography

- [1] Jane Doe. Predictive analytics in hospitals. *International Healthcare Review*, 2023.
- [2] John Smith. Artificial intelligence in clinical systems. *Journal of Medical Computing*, 2024.